

getrangefromclass Function Test Results

Test Case 1: uint8 Image

"Input:" I = uint8([1 2; 3 4]);

"Input type:" "uint8"

"Output type:" "constant"

"Range:" [0. 255.]

Test Case 2: int8 Image

"Input:" I = int8([1 -2; 3 -4]);

"Input type:" "int8"

"Output type:" "constant"

"Range:" [-128. 127.]

Test Case 3: uint16 Image

"Input:" I = uint16([1 2; 3 4]);

"Input type:" "uint16"

"Output type:" "constant"

"Range:" [0. 65535.]

Test Case 4: int16 Image

"Input:" I = int16([1 -2; 3 -4]);

"Input type:" "int16"

"Output type:" "constant"

"Range:" [-32768. 32767.]

Test Case 5: uint32 Image

"Input:" I = uint32([1 2; 3 4]);

"Input type:" "uint32"

"Output type:" "constant"

"Range:" [0. 4.295D+09]

Test Case 6: int32 Image

"Input:" I = int32([1 -2; 3 -4]);

"Input type:" "int32"

"Output type:" "constant"

"Range:" [-2.147D+09 2.147D+09]

Test Case 7: uint64 Image

"Input:" I = uint64([1 2; 3 4]);

"Input type:" "uint64"

"Output type:" "constant"

"Range:" [0. 1.845D+19]

Test Case 8: int64 Image

"Input:" I = int64([1 -2; 3 -4]);

"Input type:" "int64"

"Output type:" "constant"

"Range:" [-9.223D+18 9.223D+18]

Test Case 9: Double Image

"Input:" I = rand(5,5);

"Input type:" "constant"

"Output type:" "constant"

"Range:" [0. 1.]

Test Case 10: Boolean Image

"Input:" I = [%t %f; %f %t];

"Input type:" "boolean"

"Output type:" "constant"

"Range:" [0. 1.]

Test Case 11: Actual Grayscale Image

"Input:" I = imread("cameraman.jpeg");

"Input type:" "uint8"

"Output type:" "constant"

"Range:" [0. 255.]

Test Case 12: Empty Image

"Input:" []

"Input type:" empty matrix

"Output:" "getrangefromclass: img must be an image"

Test Case 13: String Input

"Input:" "hello"

"Input type:" string

"Output:" "getrangefromclass: unrecognized image class string"

Test Case 14: List Input

"Input:" list(1,2,3)

"Input type:" list

"Output:" "getrangefromclass: unrecognized image class list"

Test Case 15: No Input

"Input:" ()

"Input type:" missing argument

"Output:" "getrangefromclass: one input argument required."

Datatype vs Range Table

Input Type	Output Range
uint8	[0, 255]
int8	[-128, 127]
uint16	[0, 65535]
int16	[-32768, 32767]
uint32	[0, 4294967295]
int32	[-2147483648, 2147483647]
double/constant	[0, 1]
logical/boolean	[0, 1]